Constructing Character Tables: From S_3 to D_4

Week 6 · Lecture 11

Mehmet Özkan

Group Theory

Reading for Lecture 11

- **Zee, II.3, pp. 114–118:** the character-table rules and the construction for S_3.
- **Zee, II.3, pp. 125–127:** the square group D_4, its matrices, classes and character table.
- **Bünemann, §4.2, pp. 52–55; §5.2, pp. 59–64:** character tables, dimensions and reduction.
- **Bünemann, §3.2.2, pp. 32–33:** the even-dihedral class structure.

We retain the notation of W4–W5. References use printed page numbers. The optional material at the end can be omitted from the two lectures.

Group Theory

What Week 5 allows us to do

For a finite group G of order N, let C_i be its conjugacy classes and $n_i = |C_i|$. Then

$$\#\{\text{inequivalent complex irreps}\} = \#\{\text{classes}\}, \qquad \sum_\alpha d_\alpha^2 = N.$$

The characters obey

$$\frac{1}{N}\sum_i n_i \chi_\alpha(C_i)^* \chi_\beta(C_i) = \delta_{\alpha\beta}.$$

Today we use these statements to *construct* the tables. We first revisit S_3, then carry out a new construction for D_4.

Group Theory

Conventions and the books' labels

All irreducibility statements are over $\mathbb{C}$, even when we use real matrices. Vectors are columns; products act from right to left.

S_3 label in W4–W5	1	D_{sgn}	D_E
Label used here	A_1	A_2	E
Zee, II.3	1	$\bar{1}$	2

Bünemann uses A_1, A_2, E in Table 4.1. These are names for representations, not group elements. In particular, E here labels a two-dimensional irrep; the identity element is e.

Group Theory

The S_3 classes: why two rotations share a class

Keep $r = (123)$ and $s = (23)$. The relations are

$$r^3 = s^2 = e, \qquad s^{-1}rs = r^{-1} = r^2.$$

Thus r and r^2 are conjugate in S_3. Conjugating a transposition relabels its two entries, so the three transpositions are conjugate too. The classes are

$$\{e\}, \qquad \{r, r^2\}, \qquad \{s, rs, r^2s\},$$

with sizes $1, 2, 3$. In the subgroup $\langle r \rangle \cong \mathbb{Z}_3$, r and r^2 had separate classes: the conjugating element s was absent.

Textbook: Zee, II.3, pp. 116–117.

Group Theory

Finding the S_3 dimensions

Three classes mean three inequivalent irreps. One is the trivial representation, with dimension one. If the other dimensions are a, b, then

$$1 + a^2 + b^2 = 6, \qquad a, b \in \{1, 2, 3, \ldots\}.$$

The only possibility, up to their order, is $a = 1$, $b = 2$:

$$\boxed{d_{A_1} = 1, \qquad d_{A_2} = 1, \qquad d_E = 2.}$$

The dimensions are character values at e. They are not the class sizes $1, 2, 3$.

Group Theory

Deriving the two one-dimensional representations

In one dimension, write $D(r) = a$ and $D(s) = b$, where a, b are nonzero complex numbers. The group relations give

$$a^3 = 1, \qquad b^2 = 1, \qquad bab = a^{-1}.$$

Numbers commute, so $bab = ab^2 = a$. Hence $a = a^{-1}$, or $a^2 = 1$. Together with $a^3 = 1$, this forces $a = 1$.

The two choices $b = 1, -1$ give

$$\chi_{A_1} = (1, 1, 1), \qquad \chi_{A_2} = (1, 1, -1).$$

The second is precisely the sign representation of S_3.

Group Theory

The remaining row follows from orthogonality

Write the unknown two-dimensional character as $(2, x, y)$, using the same three-class order as above.

Orthogonality to the two known rows gives

$$2 + 2x + 3y = 0, \qquad 2 + 2x - 3y = 0.$$

Subtracting gives $6y = 0$. Substituting $y = 0$ into either equation gives $x = -1$. Thus

$$\boxed{\chi_E = (2, -1, 0).}$$

Its squared norm is $(4 + 2)/6 = 1$, as required. No two-dimensional matrices were needed for this calculation.

Textbook: Zee, II.3, p. 117; compare Bünemann, Table 4.1.

Group Theory

The completed $S_3 \cong D_3$ table

	e	r, r^2	s, rs, r^2s
Class size	1	2	3
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

This agrees with the matrices and traces obtained in W4. The new point is that dimensions, one-dimensional representations and orthogonality determine the table.

Our irreps are rows and classes are columns, as in Bünemann. Zee transposes this arrangement; his names for row and column orthogonality therefore interchange with ours.

Group Theory

A familiar decomposition, now read from the table

The permutation representation on three vertices has

$$\chi_{P_3} = (3, 0, 1).$$

Using $m_\alpha = N^{-1} \sum_i n_i \chi_\alpha(C_i)^* \chi_{P_3}(C_i)$,

$$m_{A_1} = \frac{3+3}{6} = 1, \quad m_{A_2} = \frac{3-3}{6} = 0, \quad m_E = \frac{2 \cdot 3}{6} = 1.$$

Hence $P_3 \simeq A_1 \oplus E$. The invariant line consists of constant vectors; the other summand is the zero-sum plane. This recovers the explicit basis change of W4.

Group Theory

Exercise: multiplicity is not dimension

An S_3 representation D has character

$$\chi_D = (5, 2, 1)$$

on classes of sizes $1, 2, 3$.

1. Find the multiplicities of A_1, A_2, E.
2. Check the dimension and reconstruct all three character values.
3. Compute $\langle \chi_D, \chi_D \rangle_G$. Does it count the dimension, the number of copies, or something else?

Group Theory

Exercise: solution

The multiplicities are

$$m_{A_1} = \frac{5+4+3}{6} = 2, \quad m_{A_2} = \frac{5+4-3}{6} = 1, \quad m_E = \frac{10-4}{6} = 1.$$

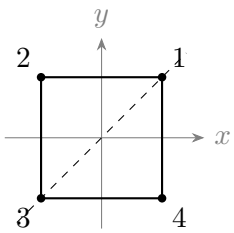
Thus $D \simeq 2A_1 \oplus A_2 \oplus E$. Its dimension is $2+1+2=5$, and its character is

$$2(1,1,1) + (1,1,-1) + (2,-1,0) = (5,2,1).$$

The squared norm is $(25+8+3)/6 = 6 = 2^2 + 1^2 + 1^2$. There are four irreducible copies in total, but the squared norm counts the sum of squared multiplicities.

Group Theory

The square: fix the geometry before the labels



Place vertices at $(a, a), (-a, a), (-a, -a), (a, -a)$, with $a > 0$.

$$r = (1234), \qquad s = (24).$$

r is a counterclockwise quarter-turn. s reflects in $y = x$, fixing vertices 1 and 3. These are the square conventions used earlier in the course.

Group Theory

The eight elements as permutations

Element	Permutation	Geometric action
e	e	identity
r	(1234)	$90°$ rotation
r^2	$(13)(24)$	$180°$ rotation
r^3	(1432)	$270°$ rotation
s	(24)	reflection in $y = x$
rs	$(12)(34)$	reflection in $x = 0$
r^2s	(13)	reflection in $y = -x$
r^3s	$(14)(23)$	reflection in $y = 0$

Group Theory

Relations and the normal form

The generators obey

$$r^4 = s^2 = e, \qquad srs = r^{-1}, \qquad sr = r^{-1}s.$$

The last relation moves each s to the right in any word. Pairs of s cancel, and powers of r reduce modulo four. Every element therefore has the form

$$r^k s^\varepsilon, \qquad k = 0, 1, 2, 3, \quad \varepsilon = 0, 1.$$

The eight permutations on the previous page are distinct, so these normal forms are unique. The group has order 8.

Group Theory

Why the mixed relation must be checked

To construct a representation, choose invertible matrices R, F satisfying

$$R^4 = I, \qquad F^2 = I, \qquad FRF = R^{-1}.$$

Then define $D(r^k s^\varepsilon) = R^k F^\varepsilon$. The multiplication rule is

$$(r^k s^\varepsilon)(r^\ell s^\eta) = r^{\,k+(-1)^\varepsilon \ell} s^{\varepsilon+\eta},$$

with exponents reduced modulo 4 and 2. The same rule holds for R, F, proving the representation property.

The two order relations alone do not specify D_4. We keep the mixed relation explicit throughout.

Group Theory

Conjugacy classes of the rotations

Use the PS2 convention $b = g^{-1}ag$ (Zee, I.2, Eq. (2), p. 58). A power of r commutes with every rotation. Since $s^{-1} = s$, reflection gives

$$s^{-1}r^k s = r^{-k}.$$

Therefore r and r^3 are conjugate, while $r^2 = r^{-2}$ is unchanged. This gives

$$\{e\}, \qquad \{r^2\}, \qquad \{r, r^3\}.$$

These are whole classes. Conjugation by either generator preserves each displayed set, and r, s generate the group.

Group Theory

Why the reflections split into two classes

As in PS2, Problem 2(a), use $sr^j = r^{-j}s$. For $g = r^j$ this gives

$$r^{-j}(r^k s)r^j = r^{k-j}sr^j = r^{k-2j}s.$$

For $g = s$, we get $s^{-1}(r^k s)s = sr^k = r^{-k}s$. Both operations preserve the parity of k. Taking $j = 1$ interchanges the two exponents of each parity modulo four, giving exactly two classes

$$\boxed{\{s, r^2 s\}, \qquad \{rs, r^3 s\}.}$$

In PS2, Problem 2(b), the first pair of axes passes through opposite atoms and the second through opposite bond midpoints.

Group Theory

The five D_4 classes and our column order

Class	Elements	Description	Size
C_1	e	identity	1
C_2	r^2	half-turn	1
C_3	r, r^3	quarter-turns	2
C_4	rs, r^3s	side-midpoint mirrors	2
C_5	s, r^2s	diagonal mirrors	2

Their sizes sum to 8. This is Zee's class order, written in our generators: his $R = r$, his $d_1 = s$, and his $\{r_x, r_y\} = \{rs, r^3s\}$.

Textbook: Zee, II.3, pp. 125–127.

Group Theory

Planar D_4 and Bünemann's point groups

Our D_4 is the order-eight symmetry group of the square in the plane, as in Zee. In three-dimensional point-group notation, the same plane operations extended with z unchanged form C_{4v}.

Bünemann's proper-rotation group D_4 instead contains four half-turns about axes in the xy plane. Restricted to that plane, those half-turns act as our reflections. Thus

$$D_4 \text{ (abstract)} \cong C_{4v} \cong D_4 \text{ (proper rotations)}.$$

The abstract tables agree after matching classes. Their actions on a spatial vector can differ: a vertical mirror preserves z, but a half-turn about an axis in the xy plane reverses it.

Textbook: Bünemann, §3.2.2, pp. 32–33, and Table 3.2, p. 41.

Group Theory

Determining the D_4 dimensions

Five classes require five inequivalent complex irreps. Their dimensions satisfy

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 = 8.$$

Subtract one from each square. The five excesses $d_i^2 - 1$ sum to $8 - 5 = 3$. Each can be $0, 3, 8, 15, \ldots$ for $d_i = 1, 2, 3, 4, \ldots$.

A dimension of three or greater already contributes at least 8. Exactly one excess must therefore be 3 and the other four must be zero. Up to order, the dimensions are

$$\boxed{1, 1, 1, 1, 2.}$$

Textbook: Zee, II.3, pp. 126–127; Bünemann, Table 4.2.

Group Theory

One-dimensional representations: start with the relations

Set $D(r) = a$ and $D(s) = b$. The three relations become

$$a^4 = 1, \qquad b^2 = 1, \qquad bab = a^{-1}.$$

Because a, b are numbers, $bab = ab^2 = a$. Therefore

$$a^2 = 1, \qquad a = \pm 1, \qquad b = \pm 1.$$

All four sign pairs satisfy all the relations and hence define representations. They are inequivalent: a one-dimensional similarity transformation cannot change a scalar.

Although r has order four, its image can have order two or one. A representation need not be faithful.

Group Theory

From two signs to a complete character row

In one dimension, the matrix and its trace are the same number. For $D(r) = a$, $D(s) = b$, we have

$$D(r^2) = a^2 = 1, \quad D(r^3) = a, \quad D(rs) = ab, \quad D(r^2 s) = b.$$

The character in our five-class order is therefore

$$\boxed{\chi_{a,b} = (1, 1, a, ab, b).}$$

The order is $(e; r^2; r, r^3; rs, r^3 s; s, r^2 s)$. We must not exchange the two reflection columns while retaining their old labels.

Group Theory

Naming the four one-dimensional rows

	a	b	e	r^2	r, r^3	rs, r^3s	s, r^2s
A_1	1	1	1	1	1	1	1
A_2	1	-1	1	1	1	-1	-1
B_1	-1	-1	1	1	-1	1	-1
B_2	-1	1	1	1	-1	-1	1

A means character $+1$ under the principal quarter-turn, and B means -1. Our subscripts are fixed by the displayed mirror columns. Later we identify B_1 with x^2-y^2 and B_2 with xy.

Textbook: For point-group naming conventions, see Bünemann, §7.1, pp. 90–91.

Group Theory

The two-dimensional plane representation

The geometric actions on a column $(x, y)^T$ are

$$r : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} -y \\ x \end{pmatrix}, \qquad s : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y \\ x \end{pmatrix}.$$

Thus choose

$$R = D_E(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad F = D_E(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives $R^2 = -I_2$, $R^4 = I_2$, $F^2 = I_2$ and $FRF = R^{-1}$. These matrices define a representation of the whole group.

Group Theory

Taking traces of the plane matrices

Besides R and F, the matrices we need are

$$R^2 = -I_2, \quad R^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad RF = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R^3F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Also $R^2F = -F$. All four reflection matrices have trace zero, as do R, R^3. Hence

$$\boxed{\chi_E = (2, -2, 0, 0, 0).}$$

This agrees with Zee's defining two-dimensional representation, with the generator labels matched as above.

Textbook: Zee, II.3, Eq. (18), p. 125.

Group Theory

Why the plane representation is irreducible

The character test gives

$$\langle \chi_E, \chi_E \rangle_{D_4} = \frac{1}{8}\left(1 \cdot 2^2 + 1 \cdot (-2)^2\right) = 1.$$

Therefore the plane representation is irreducible over $\mathbb{C}$.

For a geometric proof, any nonzero proper subspace of $\mathbb{C}^2$ is a line $L = \operatorname{span}\{v\}$. If R preserves L, then $Rv = \lambda v$. The distinct eigenvalues $i, -i$ give precisely two candidate lines.

If $Rv = iv$, then $R(Fv) = FR^{-1}v = -iFv$. The reflection F exchanges the two eigenlines, so neither is invariant under both generators.

Group Theory

The completed D_4 character table

	e	r^2	r, r^3	rs, r^3s	s, r^2s
Class size	1	1	2	2	2
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

We have constructed all five irreps, with $1 + 1 + 1 + 1 + 4 = 8$. Here C_4 and C_5 are the side and diagonal mirror classes. We check orthogonality next.

Group Theory

Checking rows: remember the class weights

For example, B_1 and B_2 have inner product

$$\langle \chi_{B_1}, \chi_{B_2} \rangle = \frac{1}{8}(1 + 1 + 2 - 2 - 2) = 0.$$

Every one-dimensional row has squared norm

$$\frac{1}{8}(1 + 1 + 2 + 2 + 2) = 1.$$

The E row is orthogonal to all four: only its first two entries contribute, giving $(2 - 2)/8 = 0$. Its squared norm was checked earlier.

The five column weights are the class sizes $(1, 1, 2, 2, 2)$.

Group Theory

A second check: column orthogonality

In our convention, column orthogonality reads

$$\sum_{\alpha} \chi_\alpha(C_i)^* \chi_\alpha(C_j) = \frac{N}{n_i} \delta_{ij}.$$

Here the sum runs over irreps. There are no class weights inside this sum.

To recall why it holds, set $U_{\alpha i} = \sqrt{n_i/N}\, \chi_\alpha(C_i)$. Row orthogonality gives $UU^\dagger = I$. Since the table is square, $U^\dagger U = I$ as well, which gives the displayed formula.

Textbook: Zee, II.3, p. 114; Bünemann, Eq. (5.10), p. 59.

Group Theory

Column checks for D_4

The identity and half-turn columns are orthogonal:

$$1 + 1 + 1 + 1 + 2(-2) = 0.$$

The squared length of a reflection column is

$$1^2 + (-1)^2 + 1^2 + (-1)^2 + 0^2 = 4 = \frac{8}{2}.$$

The side and diagonal columns have inner product

$$1 + 1 - 1 - 1 + 0 = 0.$$

The identity column has squared length 8; this is again the dimension-squared sum rule.

Group Theory

A group element is not a numerical matrix

In the plane representation, $D_E(r^2) = -I_2$. In every one-dimensional representation, however,

$$D_{A_1}(r^2) = D_{A_2}(r^2) = D_{B_1}(r^2) = D_{B_2}(r^2) = 1.$$

Thus r^2 is not an abstract "minus identity" acting as $-I$ in every representation. It is a group element whose matrix depends on the representation.

All four one-dimensional representations send r^2 to the identity matrix, so none is faithful. The geometric plane representation is faithful: its eight matrices are distinct.

Textbook: Zee's discussion after Eq. (18), pp. 125–126.

Group Theory

Exercise: recover the E row without its matrices

Suppose the four one-dimensional rows have been constructed, and the remaining row is

$$\chi_E = (2, x, y, z, w).$$

Use column orthogonality between the identity column and each of the other four columns to determine x, y, z, w.

Then check the squared norm of the resulting row. This gives a second route to the same table, using only the W5 theorems.

Group Theory

Exercise: solution

For the half-turn column, column orthogonality with the identity gives

$$4 + 2x = 0 \quad \Longrightarrow \quad x = -2.$$

For each of the other columns, the four one-dimensional entries sum to zero. Hence

$$0 + 2y = 0, \qquad 0 + 2z = 0, \qquad 0 + 2w = 0.$$

Therefore $\chi_E = (2, -2, 0, 0, 0)$. Its squared norm is $(4 + 4)/8 = 1$. The matrices supplied an explicit realization; orthogonality already fixed their traces.

Group Theory

Lecture 11 in brief

1. Determine the conjugacy classes and their sizes.
2. Use the number of classes and $\sum_\alpha d_\alpha^2 = N$ to constrain the dimensions.
3. Construct one-dimensional representations from the group relations.
4. Obtain remaining rows from known matrices or orthogonality, and check the table.

For D_4, this gives A_1, A_2, B_1, B_2, E with dimensions $1, 1, 1, 1, 2$. Next we use the table to find invariant subspaces and normal modes.

Decompositions and Normal Modes of a Square

Week 6 · Lecture 12

Mehmet Özkan

Group Theory

Reading for Lecture 12

- **Zee, II.3, pp. 118–119:** permutation characters and fixed points; pp. 125–127 for D_4.
- **Bünemann, §5.2, pp. 63–64:** multiplicities and the irreducibility test.
- **Zee, III.2, pp. 168–170:** harmonic motion, symmetry and degeneracy. Our scalar square model is a worked extension of this method.
- **Bünemann, §6.3.2, p. 81, and §7.1, pp. 89–91:** invariant eigenspaces and representation functions.

The physical example requires only Newton's equation and matrix eigenvectors. No quantum mechanics is assumed.

Group Theory

Two questions about a reducible representation

For a representation D, we can ask:

1. **Which irreps occur, and how often?** Characters answer this through

$$m_\alpha = \frac{1}{N} \sum_i n_i \chi_\alpha(C_i)^* \chi_D(C_i).$$

2. **Which vectors span the irreducible pieces?** We must then construct an appropriate basis, using geometry, direct calculation or projection operators.

We will do both for the action on the four vertices of a square.

Group Theory

Why a permutation character counts fixed vertices

Let e_j be the standard basis vector associated with vertex j. Define

$$P(g)e_j = e_{g(j)}, \qquad P(g)_{ij} = \delta_{i,g(j)}.$$

The diagonal entry $P(g)_{jj}$ is 1 exactly when $g(j) = j$, and otherwise it is 0. Therefore

$$\boxed{\chi_P(g) = \operatorname{tr} P(g) = \#\{j : g(j) = j\}.}$$

This counts vertices individually left in place. It does not count the number of cycles in the permutation, or whether the entire square is preserved.

Textbook: Zee, II.3, pp. 118–119.

Group Theory

The four-dimensional permutation matrices

For $r = (1234)$ and $s = (24)$, the column rule gives

$$P(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad P(s) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Thus $P(r)(q_1, q_2, q_3, q_4)^T = (q_4, q_1, q_2, q_3)^T$. The value originally at vertex 1 moves to vertex 2.

The matrices act on four vertex amplitudes, not on the two Cartesian components of a point. They differ from R, F in Lecture 11.

Group Theory

The character of the vertex representation

Class	e	r^2	r, r^3	rs, r^3s	s, r^2s
Class size	1	1	2	2	2
Fixed vertices	4	0	0	0	2

The side-midpoint mirrors exchange two pairs of vertices, leaving none fixed. Each diagonal mirror fixes its two vertices on the mirror line and exchanges the other two.

$$\boxed{\chi_P = (4, 0, 0, 0, 2).}$$

We have obtained all character values without constructing all eight 4×4 matrices.

Group Theory

Computing all five multiplicities

Only the identity and diagonal-reflection columns contribute:

$$m_\alpha = \frac{1}{8}\left[4\chi_\alpha(e) + 4\chi_\alpha(C_5)\right].$$

All entries are real in this table. Substituting the five rows gives

$$\begin{aligned} m_{A_1} &= (4+4)/8 = 1, \quad m_{A_2} = (4-4)/8 = 0, \\ m_{B_1} &= (4-4)/8 = 0, \quad m_{B_2} = (4+4)/8 = 1, \\ m_E &= (8+0)/8 = 1. \end{aligned}$$

Hence $\boxed{P \simeq A_1 \oplus B_2 \oplus E}$ in our stated geometry and label convention.

Group Theory

Three checks on the decomposition

Dimension:

$$4 = 1 + 1 + 2.$$

Character on every class:

$$(1, 1, 1, 1, 1) + (1, 1, -1, -1, 1) + (2, -2, 0, 0, 0) = (4, 0, 0, 0, 2).$$

Squared norm:

$$\langle \chi_P, \chi_P \rangle = \frac{1}{8}(16 + 2 \cdot 4) = 3 = 1^2 + 1^2 + 1^2.$$

There are three inequivalent pieces, each occurring once. We now find their basis vectors.

Group Theory

The two one-dimensional invariant subspaces

Consider the normalized vectors

$$u_A = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}, \qquad u_B = \frac{1}{2}\begin{pmatrix}1\\-1\\1\\-1\end{pmatrix}.$$

The generator matrices give

$$P(r)u_A = u_A, \quad P(s)u_A = u_A; \qquad P(r)u_B = -u_B, \quad P(s)u_B = u_B.$$

Thus u_A spans A_1, while u_B spans B_2. The B_2 line is invariant even though its individual vectors change sign under r.

Group Theory

A basis for the remaining plane

Choose

$$u_x = \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \qquad u_y = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}.$$

Their entries follow the signs of the square vertices' x and y coordinates. All four vectors u_A, u_B, u_x, u_y have unit norm and are mutually orthogonal.

For example, $u_x^T u_y = (1 - 1 + 1 - 1)/4 = 0$. Therefore they form an orthonormal basis of the four-dimensional space.

Group Theory

Check how both generators act on the plane

Using the explicit permutations of components,

$$P(r)u_x = u_y, \qquad P(r)u_y = -u_x,$$

and

$$P(s)u_x = u_y, \qquad P(s)u_y = u_x.$$

Therefore $\mathrm{span}\{u_x, u_y\}$ is invariant under both generators and hence under every group element. In the ordered basis (u_x, u_y), their matrices are

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = R, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = F.$$

This identifies the plane as the same irrep E used in Lecture 11.

Group Theory

Why these vectors become columns of S

Write a general vector as

$$q = a_A u_A + a_B u_B + a_x u_x + a_y u_y = Sa,$$

where $a = (a_A, a_B, a_x, a_y)^T$. The coefficient of a_A in Sa is the first column, so that column must be u_A, and similarly for the others:

$$S = (u_A\ u_B\ u_x\ u_y) = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}.$$

Orthonormality gives $S^T S = I_4$, so $a = S^T q$.

Group Theory

The same basis makes every matrix block diagonal

Since $q' = P(g)q$ and $q = Sa$, the new coordinates obey

$$a' = S^T P(g) S a.$$

For the generators, the actions just calculated give

$$S^T P(r) S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad S^T P(s) S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The blocks are A_1, B_2, E. Because r, s generate the group, this one common basis works for every $P(g)$.

Group Theory

Extracting the four coordinates

Multiplying $a = S^T q$ gives

$$\begin{aligned} a_A &= \tfrac{1}{2}(q_1 + q_2 + q_3 + q_4), \\ a_B &= \tfrac{1}{2}(q_1 - q_2 + q_3 - q_4), \\ a_x &= \tfrac{1}{2}(q_1 - q_2 - q_3 + q_4), \\ a_y &= \tfrac{1}{2}(q_1 + q_2 - q_3 - q_4). \end{aligned}$$

These are coordinates along the new basis vectors, not four new vertices. The original vector is recovered from

$$q = a_A u_A + a_B u_B + a_x u_x + a_y u_y.$$

Group Theory

Exercise: resolve a displacement into its pieces

Let $q = q_* \, (3, -1, 1, 1)^T$, where q_* is a fixed amplitude.

1. Find a_A, a_B, a_x, a_y.
2. Write the A_1, B_2 and E components separately.
3. Predict the coefficients of $P(r)q$ using the block matrices, then check them by permuting the original components.

The E component has two coordinates, but it is one irreducible two-dimensional subspace.

Group Theory

Exercise: solution

The coordinates are $(a_A, a_B, a_x, a_y) = q_*(2, 2, 2, 0)$. Hence

$$q = 2q_* u_A + 2q_* u_B + 2q_* u_x.$$

The separate components are $q_*(1, 1, 1, 1)^T$, $q_*(1, -1, 1, -1)^T$, and $q_*(1, -1, -1, 1)^T$. Under r, the coefficients become $q_*(2, -2, 0, 2)$. Reconstructing gives

$$P(r)q = q_*(1, 3, -1, 1)^T,$$

which agrees with $(q_4, q_1, q_2, q_3)^T$.

Group Theory

A physical realization: four scalar oscillators

Attach one real amplitude $q_j(t)$ to each vertex, with four equal inertial parameters $M > 0$. Under a square symmetry, the amplitudes are permuted by $P(g)$.

One realization uses four oscillators constrained to move vertically, with identical restoring and coupling terms. Use rotations about z and vertical mirror planes: these operations preserve the direction of positive vertical displacement.

There are **four** coordinates. This is a scalar-oscillator model, not a free square molecule with two in-plane displacement components at each vertex. Its representation is exactly the P just decomposed.

Group Theory

From Newton's equation to normal modes

For a quadratic potential, write

$$V(q) = \frac{M}{2} q^T H q, \qquad H^T = H.$$

Newton's equation is $\ddot{q} = -Hq$. To look for a mode of fixed shape u, set $q(t) = u\cos(\omega t + \phi)$. Substitution gives

$$-\omega^2 u = -Hu \quad \Longrightarrow \quad \boxed{Hu = \omega^2 u.}$$

The vector u gives relative amplitudes; ω is the angular frequency. An eigenvalue of H has units of frequency squared.

Textbook: Zee, III.2, pp. 168–169.

Group Theory

Why symmetry makes H commute with $P(g)$

Invariance of the potential means $V(P(g)q) = V(q)$ for every q. Therefore

$$q^T P(g)^T H P(g) q = q^T H q \qquad \text{for every } q.$$

The two symmetric matrices must therefore be equal. Write $P = P(g)$ for brevity: $P^T H P = H$. Since $PP^T = I$, multiplying on the left by P gives

$$\boxed{HP(g) = P(g)H.}$$

If $Hu = \lambda u$, this implies $H(P(g)u) = \lambda P(g)u$: every eigenspace of H is invariant under the symmetry.

Textbook: Compare Zee, III.2, pp. 169–170; Bünemann, §6.3.2.

Group Theory

What Schur's lemma tells us here

We found $P \simeq A_1 \oplus B_2 \oplus E$, with each irrep appearing once. A commuting H cannot mix inequivalent irreps, and it is scalar on each one. Hence

$$S^T H S = \mathrm{diag}(\lambda_A, \lambda_B, \lambda_E, \lambda_E).$$

The two E modes have the same frequency squared. The two one-dimensional modes need not share that eigenvalue or each other's.

If an irrep occurred several times, H could mix its equivalent copies. One would first diagonalize the matrix acting on those copies. Our square representation has no such repetition.

Group Theory

The general D_4-invariant dynamical matrix

Quarter-turn symmetry gives $H_{ij} = H_{r(i),r(j)}$. In particular $H_{41} = H_{12}$, since $r(4) = 1$ and $r(1) = 2$. Since $H^T = H$, we also have $H_{14} = H_{41}$.

Write the first row as $(\alpha, \beta, \gamma, \beta)$. Successive rotations give

$$H = \begin{pmatrix} \alpha & \beta & \gamma & \beta \\ \beta & \alpha & \beta & \gamma \\ \gamma & \beta & \alpha & \beta \\ \beta & \gamma & \beta & \alpha \end{pmatrix}.$$

Thus $C_4 = \langle r \rangle$ already gives this form for a real symmetric H. It also commutes with $P(s)$. The optional pages explain the resulting degeneracy in C_4 language.

Group Theory

Finding its eigenvalues using the symmetry basis

Multiplying H by the four known vectors gives

$$\begin{aligned} Hu_A &= (\alpha + 2\beta + \gamma)u_A, \\ Hu_B &= (\alpha - 2\beta + \gamma)u_B, \\ Hu_x &= (\alpha - \gamma)u_x, \qquad Hu_y = (\alpha - \gamma)u_y. \end{aligned}$$

For example, the first component of Hu_x is

$$\tfrac{1}{2}(\alpha - \beta - \gamma + \beta) = \tfrac{1}{2}(\alpha - \gamma).$$

Symmetry provided the eigenvectors in advance. We only needed to apply H, not solve a fourth-degree characteristic equation.

Group Theory

Specify a model to obtain numerical frequencies

Take the potential

$$V = \frac{k_0}{2}\sum_{j=1}^{4} q_j^2 + \frac{k}{2}\sum_{j=1}^{4}(q_j - q_{j+1})^2 + \frac{\kappa}{2}\left[(q_1 - q_3)^2 + (q_2 - q_4)^2\right], \qquad q_5 = q_1.$$

Here $k_0 \geq 0$ is the local restoring constant, $k > 0$ couples adjacent sites, and $\kappa \geq 0$ couples opposite sites. This quadratic model defines the couplings; we need not infer them from a drawing of ordinary horizontal springs.

Every term is carried into a term with the same coefficient by D_4.

Group Theory

Deriving the dynamical matrix from the potential

For vertex 1, differentiation gives

$$M\ddot{q}_1 = -k_0 q_1 - k(q_1 - q_2) - k(q_1 - q_4) - \kappa(q_1 - q_3).$$

The other equations follow by cyclically relabelling the vertices. Thus

$$H = \frac{1}{M}\begin{pmatrix} k_0 + 2k + \kappa & -k & -\kappa & -k \\ -k & k_0 + 2k + \kappa & -k & -\kappa \\ -\kappa & -k & k_0 + 2k + \kappa & -k \\ -k & -\kappa & -k & k_0 + 2k + \kappa \end{pmatrix}.$$

So $\alpha = (k_0 + 2k + \kappa)/M$, $\beta = -k/M$, and $\gamma = -\kappa/M$.

Group Theory

The frequencies and their symmetry labels

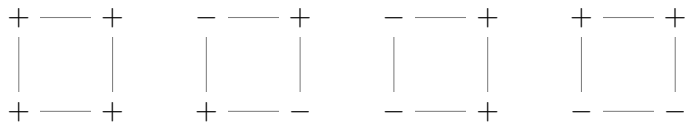
Irrep	Mode shape	Frequency squared
A_1	u_A	$\omega_A^2 = k_0/M$
B_2	u_B	$\omega_B^2 = (k_0 + 4k)/M$
E	u_x, u_y	$\omega_E^2 = (k_0 + 2k + 2\kappa)/M$

The E eigenvalue occurs twice. Any linear combination of u_x, u_y is also an eigenvector with that same eigenvalue.

The character calculation predicted the irreducible content. The potential supplied the three frequency values.

Group Theory

Reading the four mode shapes



Signs give the vertical displacement at each labelled vertex, with common magnitude $1/2$. They do not depict motion along the square's edges.

u_A moves all sites together; u_B alternates around the square. The pair u_x, u_y spans the degenerate E space. Rotating u_x gives u_y, then $-u_x$.

Group Theory

Zero modes and coincident frequencies

If $k_0 = 0$, a uniform shift changes no difference $q_i - q_j$, so $Hu_A = 0$. Its time dependence satisfies $\ddot{a}_A = 0$:

$$a_A(t) = A + Bt.$$

This zero frequency follows from the specified potential's uniform-shift symmetry. Membership in A_1 alone would not force it.

If $\kappa = k$, the B_2 and E eigenvalues coincide. All six pairs of sites then have equal coupling, enlarging the site-permutation symmetry to S_4. Equality of these three eigenvalues is not required by D_4 alone.

Group Theory

Exercise: frequencies without a determinant

Set $\kappa = 0$ and $k_0 = k > 0$ in the scalar square model.

1. List all four frequencies, including their multiplicities.
2. Which displacement pattern has the lowest frequency?
3. Increase k_0 by $\Delta k_0 > 0$, keeping the other parameters fixed. What changes in the mode shapes and frequency squares?
4. Does symmetry determine that A_1 is always the lowest mode for an arbitrary D_4-invariant matrix H?

Group Theory

Exercise: solution

The frequencies are

$$\sqrt{\frac{k}{M}} \quad (A_1), \qquad \sqrt{\frac{3k}{M}} \quad (E, \text{ twice}), \qquad \sqrt{\frac{5k}{M}} \quad (B_2).$$

The uniform mode u_A is lowest in this model. Increasing k_0 adds $(\Delta k_0/M)I_4$ to H, so all frequency squares increase by $\Delta k_0/M$ and every mode shape is unchanged.

For a general square-symmetric H, the numbers $\alpha + 2\beta + \gamma$, $\alpha - 2\beta + \gamma$ and $\alpha - \gamma$ need not have this ordering. Symmetry fixes the block structure, not the ordering of eigenvalues.

Group Theory

Towards W7: representations carried by functions

Let $M(g)$ be the 2×2 geometric plane matrix. Define the action on a function by

$$[U(g)f](\mathbf{x}) = f(M(g)^{-1}\mathbf{x}), \qquad \mathbf{x} = (x, y)^T.$$

The inverse makes this a representation:

$$[U(g)U(h)f](\mathbf{x}) = f(M(h)^{-1}M(g)^{-1}\mathbf{x}) = [U(gh)f](\mathbf{x}).$$

For our generators, $U(r)f(x, y) = f(y, -x)$ and $U(s)f(x, y) = f(y, x)$. We can now identify irreps by how polynomial functions transform.

Textbook: Representation spaces: Bünemann, §6.1; basis functions in tables: §7.1.

Group Theory

Why $x^2 - y^2$ is B_1, while xy is B_2

Set $f = x^2 - y^2$ and $h = xy$. Substitution gives

$$U(r)f = y^2 - (-x)^2 = -f, \quad U(s)f = y^2 - x^2 = -f,$$
$$U(r)h = y(-x) = -h, \qquad U(s)h = yx = h.$$

Thus their generator signs (a, b) are $(-1, -1)$ and $(-1, 1)$, respectively: f spans B_1 and h spans B_2.

Similarly, $x^2 + y^2$ spans A_1. The pair (x, y) spans E, because $U(r)x = y$, $U(r)y = -x$, $U(s)x = y$, $U(s)y = x$.

Group Theory

Evaluating a polynomial at the four vertices

Write v_j for the position of vertex j. Given a polynomial f, define the column

$$\mathrm{ev}(f) = \big(f(v_1), f(v_2), f(v_3), f(v_4)\big)^T.$$

Since $M(g)^{-1} v_j = v_{g^{-1}(j)}$, its jth entry transforms as

$$[\mathrm{ev}(U(g)f)]_j = f(v_{g^{-1}(j)}) = [P(g)\mathrm{ev}(f)]_j.$$

This proves $\mathrm{ev}\, U(g) = P(g)\, \mathrm{ev}$. In the language of W5, evaluation is an **intertwiner** from functions on the plane to amplitudes on the vertices.

Group Theory

The same irrep can have different realizations

Irrep	Polynomial f	Vertex values $\mathrm{ev}(f)$
A_1	$x^2 + y^2$	$4a^2u_A$
B_2	xy	$2a^2u_B$
E	$x,\ y$	$2au_x,\ 2au_y$
B_1	$x^2 - y^2$	0

Every vertex has $x^2 = y^2 = a^2$, so $x^2 - y^2$ vanishes at all four vertices. Schur's lemma also forces this zero. Each component of $\mathrm{ev}|_{B_1}$ maps B_1 into one of the inequivalent irreps A_1, B_2, E of P.

Group Theory

What to be able to do after Week 6

- Construct the S_3 and D_4 tables, showing where their rows and class weights come from.
- Use characters to find multiplicities, then check dimensions and reconstruct the character.
- Build a common basis that displays the invariant pieces of the square's vertex representation.
- Explain which degeneracies follow from symmetry and which frequencies require a dynamical model.

Next week: point groups and their physical representations, using the same methods to read basis functions and symmetry labels.

Optional Extensions to Week 6

Week 6 · Lecture 11–12

Mehmet Özkan

Group Theory

Optional reading and scope

- **Bünemann, §7.1, pp. 91–92** discusses the two conjugate C_4 representations and the degeneracy of a real Hamiltonian. We use the square oscillators to examine this example in classical mechanics.
- **Bünemann, §6.2, pp. 77–79** gives the matrix-element projectors in Eq. (6.14) and the character projector in Eq. (6.16).
- **Zee, IV.i1, pp. 289–291:** decomposing products using characters. Zee uses the tetrahedral group; we apply the same idea to D_4 and S_3.

These pages extend the main lectures. The full tensor-product development remains for the later angular-momentum part of the course.

Group Theory

Why quarter-turn symmetry suffices for the doublet

Let $T = P(r)$. On the E plane, $Tu_x = u_y$ and $Tu_y = -u_x$. For $\sigma = \pm 1$ define

$$v_\sigma = (u_x - i\sigma u_y)/\sqrt{2}, \qquad Tv_\sigma = i\sigma v_\sigma, \qquad v_- = v_+^*.$$

Each spans a one-dimensional complex irrep of C_4. The other two vertex modes have T eigenvalues $1, -1$, so each $i\sigma$ eigenline occurs just once.

Since $HT = TH$, write $Hv_+ = \lambda v_+$. Symmetry of the real matrix H gives $\lambda \in \mathbb{R}$. Taking the complex conjugate gives $Hv_- = \lambda v_-$. The two modes therefore have the same frequency squared.

Textbook: Compare Bünemann, §7.1, Table 7.2 and Eq. (7.3), pp. 91–92.

Group Theory

A gyroscopic coupling preserves quarter-turns

Keep the stiffness $K = MH$ and add a force linear in velocity. With $\Omega > 0$ of dimension inverse time, choose

$$M\ddot{q} = -Kq + G\dot{q}, \qquad G = \frac{M\Omega}{2}(T - T^T).$$

Here $T^T = T^{-1}$, so $G^T = -G$ and G commutes with T. The force does zero work per unit time because $\dot{q}^T G\dot{q} = 0$.

A mirror exchanges T and T^{-1}, reversing the sign of G. Under time reversal $\dot{q}$ also changes sign. With the coupling held fixed, the equation retains C_4 symmetry and breaks both mirror and time-reversal symmetry.

Group Theory

The clockwise and counterclockwise frequencies

On the doublet $Kv_\sigma = M\omega_E^2 v_\sigma$ and $Gv_\sigma = i\sigma M\Omega v_\sigma$. Substituting $q = v_\sigma e^{i\omega t}$ gives

$$-\omega^2 = -\omega_E^2 - \sigma\Omega\omega, \qquad \boxed{\omega_\sigma = \sqrt{\omega_E^2 + \Omega^2/4} + \sigma\Omega/2}.$$

We chose the positive-frequency root. The physical motion is the real part, proportional to $u_x \cos(\omega_\sigma t) + \sigma u_y \sin(\omega_\sigma t)$.

In a quarter period the pattern moves from u_x to σu_y. The $+$ wave circulates counterclockwise and the $-$ wave clockwise, with splitting $\omega_+ - \omega_- = \Omega$. Compare PS2, Problem 1(e).

Group Theory

Exercise on vertical motion under proper D_4

Keep the four vertical oscillators. Now realize r as a quarter-turn about z and s as a half-turn about the in-plane diagonal through vertices 1 and 3. A positive z displacement changes sign under this s.

1. Find the sign $\epsilon(g)$ of the transformed vertical direction on each class. Identify the irrep carried by ϵ.
2. The new action is $D_z(g) = \epsilon(g)P(g)$. Starting from $\chi_P = (4, 0, 0, 0, 2)$, compute its character and decompose it.
3. Identify the new symmetry labels of $u_A, u_B, (u_x, u_y)$. What frequencies do the same masses and couplings give?

Textbook: Bünemann, §3.2.2, pp. 32–33, and Table 3.2, p. 41.

Group Theory

Solution for the vertical amplitudes

The turns about z preserve the vertical direction. The four in-plane half-turns reverse it. Thus $\epsilon = (1, 1, 1, -1, -1) = \chi_{A_2}$ and

$$\chi_{D_z} = \epsilon\, \chi_P = (4, 0, 0, 0, -2).$$

Only e and C_5 contribute, giving $m_\alpha = (d_\alpha - \chi_\alpha(C_5))/2$. The five multiplicities $(0, 1, 1, 0, 1)$ yield

$$D_z \simeq A_2 \oplus B_1 \oplus E.$$

u_A now carries A_2, u_B carries B_1, and the pair still carries E. Since $\epsilon(g)^2 = 1$, $D_z(g)^T H D_z(g) = P(g)^T H P(g) = H$. The specified H gives the same three frequency values.

Group Theory

Optional: a projector for each irreducible type

For a unitary representation D and an irreducible character χ_α, define

$$\boxed{\Pi_\alpha = \frac{d_\alpha}{N} \sum_{g \in G} \chi_\alpha(g)^* D(g).}$$

In $D(h)^{-1} \Pi_\alpha D(h)$ each $D(g)$ becomes $D(h^{-1}gh)$. The relabelling $g \mapsto h^{-1}gh$ permutes the sum and preserves $\chi_\alpha(g)$. Thus Π_α commutes with every $D(h)$.

We will show that its image is the sum of *all copies* of irrep α. This is called the α-isotypic subspace. If the multiplicity is one, it is just that irreducible subspace.

Group Theory

Optional: why the formula is a projector

Use an orthonormal basis in which D is a direct sum of irreps. On each copy of an irrep β, the sum is block diagonal and commutes with $D^{(\beta)}$. Schur's lemma gives

$$\Pi_\alpha|_{V_\beta} = c_{\alpha\beta} I_{d_\beta}.$$

Taking the trace determines its scalar:

$$c_{\alpha\beta} d_\beta = \frac{d_\alpha}{N} \sum_g \chi_\alpha(g)^* \chi_\beta(g) = d_\alpha \delta_{\alpha\beta}.$$

So it is the identity on every α copy and zero on all other irreps. Hence $\Pi_\alpha^2 = \Pi_\alpha$, $\Pi_\alpha^\dagger = \Pi_\alpha$, and $\operatorname{rank} \Pi_\alpha = m_\alpha d_\alpha$.

Group Theory

Optional: the square's three nonzero projectors

Our orthonormal basis gives

$$\Pi_{A_1} = u_A u_A^T, \qquad \Pi_{B_2} = u_B u_B^T, \qquad \Pi_E = u_x u_x^T + u_y u_y^T.$$

For example,

$$\Pi_E = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} = \frac{1}{2}[I_4 - P(r^2)].$$

The second expression follows directly from the character sum, since χ_E is nonzero only at e, r^2. The other two projectors, Π_{A_2} and Π_{B_1}, vanish in this representation.

Group Theory

Optional: projecting one localized amplitude

Let $e_1 = (1,0,0,0)^T$. Then

$$\Pi_{A_1} e_1 = \frac{1}{4}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}, \qquad \Pi_{B_2} e_1 = \frac{1}{4}\begin{pmatrix}1\\-1\\1\\-1\end{pmatrix}, \qquad \Pi_E e_1 = \frac{1}{2}\begin{pmatrix}1\\0\\-1\\0\end{pmatrix}.$$

These three components sum to e_1. The starting vector e_1 gives the direction $u_x + u_y$ in the E plane. Applying the same projector to e_2 gives $\Pi_E e_2 = (-u_x + u_y)/2$, the perpendicular direction.

Group Theory

Optional: what a product representation acts on

If D acts on a space with basis e_i and F on one with basis f_j, their tensor-product space has basis $e_i \otimes f_j$ and dimension $(\dim D)(\dim F)$.

The symbol $\otimes$ is bilinear: for example, $(u+v)\otimes w = u\otimes w + v\otimes w$. Define

$$[D \otimes F](g)(u \otimes v) = D(g)u \otimes F(g)v.$$

Applying h then g proves the group law on these basis vectors and hence on their linear combinations. This is a representation of the *same* group acting on both factors.

Group Theory

Optional: why product characters multiply

In the product basis, matrix entries are

$$[D(g) \otimes F(g)]_{(i,j),(k,\ell)} = D(g)_{ik} F(g)_{j\ell}.$$

Taking the trace gives

$$\chi_{D\otimes F}(g) = \sum_{i,j} D(g)_{ii} F(g)_{jj} = \chi_D(g)\chi_F(g).$$

Thus we multiply characters *entry by entry*, then use the same multiplicity formula as before. Direct sums add characters; tensor products multiply them.

Textbook: Method illustrated in Zee, IV.i1, pp. 290–291.

Group Theory

Optional: decomposing $E \otimes E$ for D_4

Since $\chi_E = (2, -2, 0, 0, 0)$, the product character is

$$\chi_{E\otimes E} = (4, 4, 0, 0, 0).$$

Each one-dimensional row has value 1 at both e and r^2, so its multiplicity is $(4 + 4)/8 = 1$. For E,

$$m_E = \frac{1}{8}(2 \cdot 4 + (-2) \cdot 4) = 0.$$

Therefore

$$\boxed{E \otimes E \simeq A_1 \oplus A_2 \oplus B_1 \oplus B_2.}$$

The dimension check is $2 \cdot 2 = 1 + 1 + 1 + 1$.

Group Theory

Optional: exhibit the four product basis vectors

Let e_x, e_y be the plane basis, with $re_x = e_y$, $re_y = -e_x$, $se_x = e_y$, $se_y = e_x$. In $E \otimes E$, define

	Basis vector, up to normalization	r	s
A_1	$e_x \otimes e_x + e_y \otimes e_y$	$+$	$+$
A_2	$e_x \otimes e_y - e_y \otimes e_x$	$+$	$-$
B_1	$e_x \otimes e_x - e_y \otimes e_y$	$-$	$-$
B_2	$e_x \otimes e_y + e_y \otimes e_x$	$-$	$+$

For example, $r(e_x \otimes e_y - e_y \otimes e_x) = -e_y \otimes e_x + e_x \otimes e_y$. Applying s reverses its sign. This verifies the A_2 assignment directly.

Group Theory

Optional exercise: compare with S_3

For S_3, $\chi_E = (2, -1, 0)$ on classes of sizes $(1, 2, 3)$. Find $E \otimes E$ using its character $(4, 1, 0)$.

The multiplicities are

$$m_{A_1} = \frac{4+2}{6} = 1, \qquad m_{A_2} = \frac{4+2}{6} = 1, \qquad m_E = \frac{8-2}{6} = 1.$$

Thus $E \otimes E \simeq A_1 \oplus A_2 \oplus E$ for S_3, whereas for D_4 it splits into four one-dimensional pieces.

The dimension of a representation alone does not determine its product decomposition; the group and its character table matter.